

THEORY OF COMPUTATION (CS302)

Context-Free Grammars (CFG)

Learning Objectives

Understand Context-Free Grammars, production rules, derivations, parse trees, ambiguity, and their applications in language design.

Introduction to Context-Free Grammars

A Context-Free Grammar (CFG) is a formal grammar used to generate context-free languages. CFGs are widely used in compiler design and programming language specification.

Definition of CFG

A CFG is represented by the tuple $G = (V, T, P, S)$, where V is a set of variables, T is a set of terminals, P is a set of production rules, and S is the start symbol.

Components of CFG

- Variables (Non-Terminals)
- Terminals
- Production Rules
- Start Symbol

Production Rules

Production rules define how variables can be replaced by terminals and other variables to generate strings.

Example Grammar

$S \rightarrow aSb$
 $S \rightarrow ab$

This grammar generates strings such as: ab , $aabb$, $aaabbb$

Derivations

A derivation is a sequence of rule applications used to generate a string from the start symbol.

Leftmost Derivation

In leftmost derivation, the leftmost non-terminal is replaced first at each step.

Rightmost Derivation

In rightmost derivation, the rightmost non-terminal is replaced first at each step.

Parse Trees

A parse tree is a tree representation showing how a string is derived from a grammar.

Ambiguous Grammar

A grammar is ambiguous if a string can have more than one parse tree or derivation.

Applications of CFG

- Compiler Design
- Syntax Analysis
- Programming Languages
- Natural Language Processing

Solved Example

Grammar: $S \rightarrow aSb \mid ab$

Generate 'aabb'

$S \Rightarrow aSb$

$\Rightarrow a(ab)b$

$\Rightarrow aabb$

Exam Important Questions

1. Define CFG.
2. Explain components of CFG.
3. Differentiate terminals and non-terminals.
4. Explain leftmost and rightmost derivations.
5. What is ambiguity in CFG?
6. Draw a parse tree for a given grammar.

Summary

Context-Free Grammars provide a structured way to describe programming language syntax and context-free languages. They are essential for parsing and compiler construction.